SOVEREIGN GOVERNMENT CLOUD

CSIR Secure Web Appliance OS (CSIR-SWA OS)

Next-Generation Hardened, Immutable, and Containerized Web Hosting Appliance

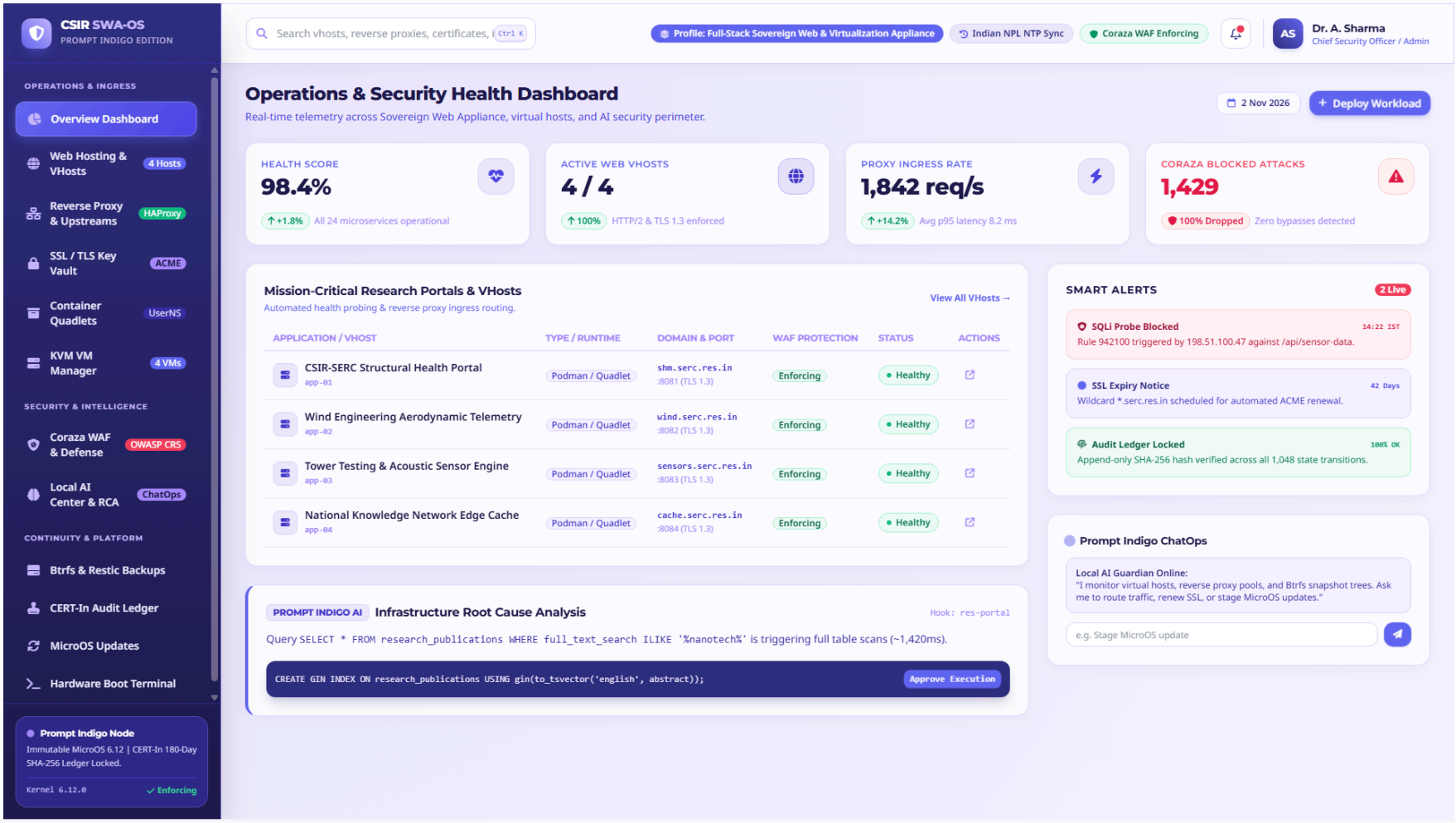
CORE ARCHITECTURAL MANDATE

- Engineered for classified research laboratories, government web portals, and critical public infrastructure.
- Prompt Indigo AI SaaS Architecture with Tailwind CSS, daisyUI, and Glassmorphic aesthetics.
- Integrated Multi-Tenant Web Hosting, Virtual Host Manager, KVM VM Manager, HAProxy Proxy, and ACME SSL Key Vault.

VISUAL QUALIFICATION: WEB CONSOLE SUBSYSTEM

Figure 1: Central Operations & Security Health Overview Dashboard

Dedicated Verification Slide



Executive Health KPI
98.4% operational health score across 24 microservices

Proxy Ingress Telemetry
1,842 req/s throughput with sub-10ms p95 latency

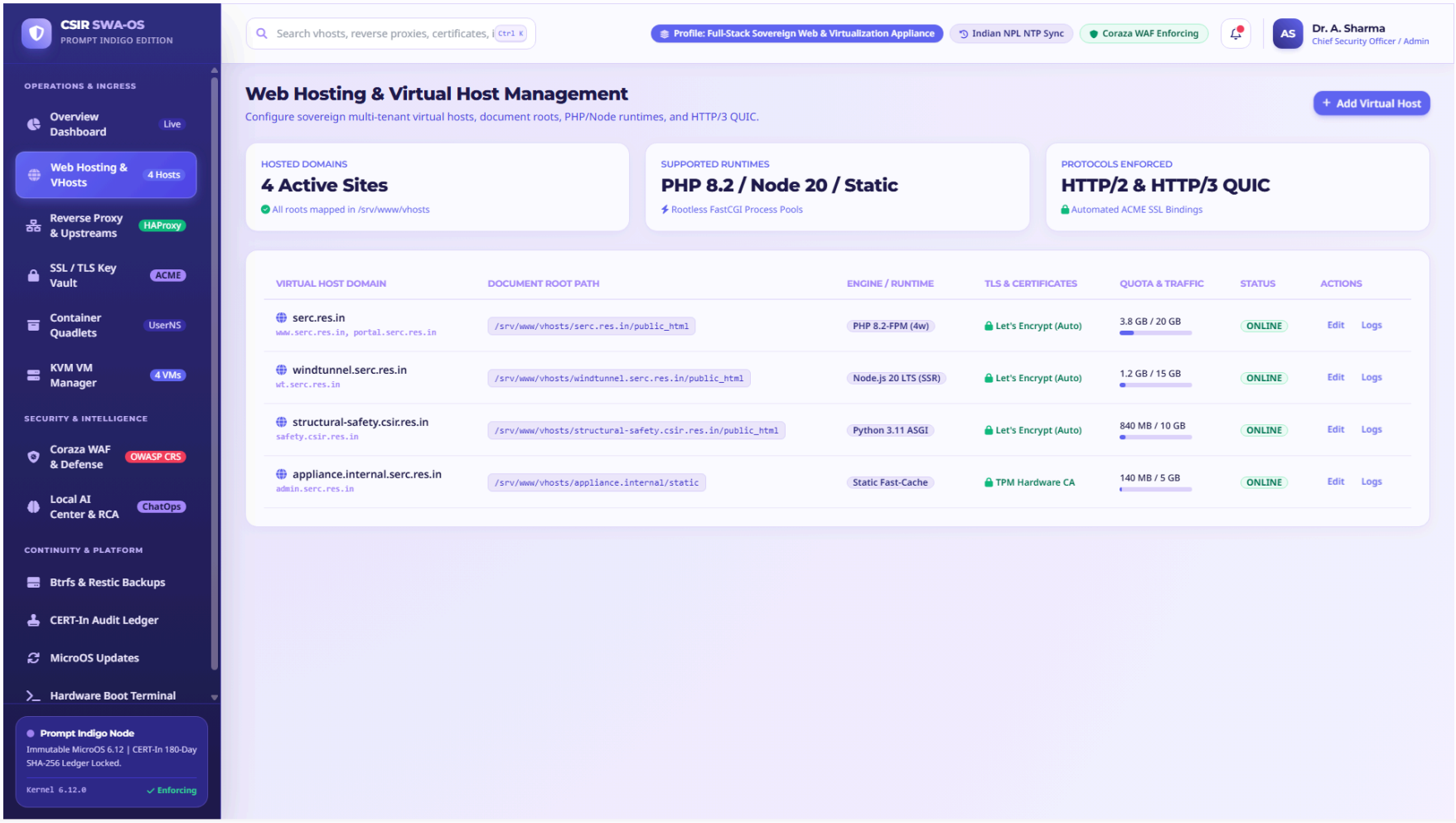
Coraza Threat Shield
1,429 blocked attacks with 100% drop & zero bypasses

Prompt Indigo ChatOps
Real-time security timeline with sovereign AI assistant

VISUAL QUALIFICATION: WEB CONSOLE SUBSYSTEM

Figure 2: Native Web Hosting & Multi-Tenant Virtual Host Fleet

Dedicated Verification Slide



Multi-Tenant Isolation
Enforced document roots under /srv/www/vhosts with strict POSIX perms

Polyglot Runtimes
Native support for PHP 8.2-FPM, Node.js 20 SSR, Python ASGI & Static

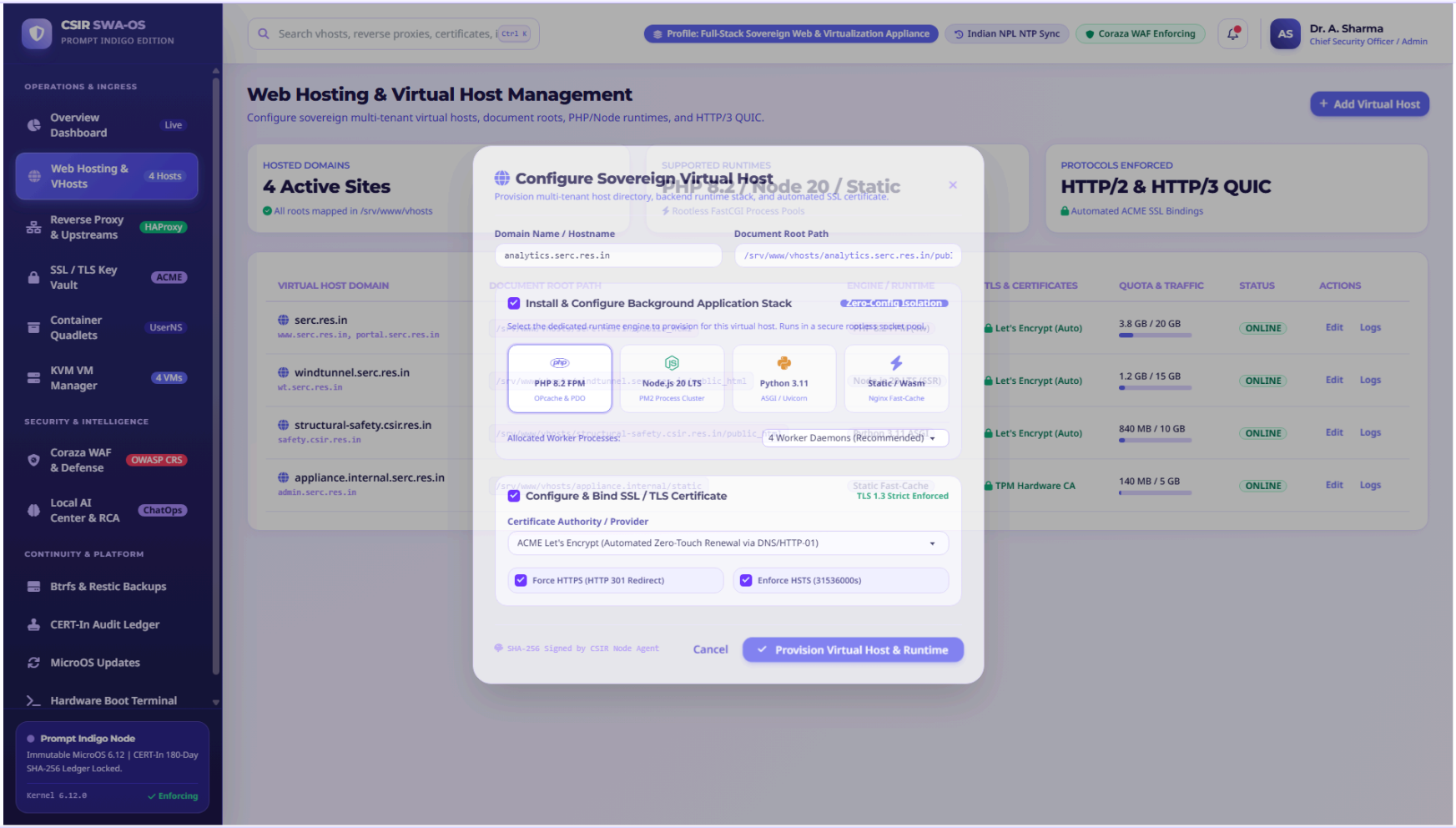
Storage Quota Bars
Visual disk consumption tracking with automated threshold enforcement

Active SSL Binding
Instant binding to automated Let's Encrypt and wildcard certificates

VISUAL QUALIFICATION: WEB CONSOLE SUBSYSTEM

Figure 3: Virtual Host Configuration & Background Stack Setup Modal

Dedicated Verification Slide



Application Runtime Selection
Interactive stack choice for PHP 8.2-FPM, Node.js 20 SSR, Python 3.11 ASGI, or Static

Tunable Worker Daemons
Selectable 2, 4, 8, or 16 isolated worker processes running in unprivileged cgroups

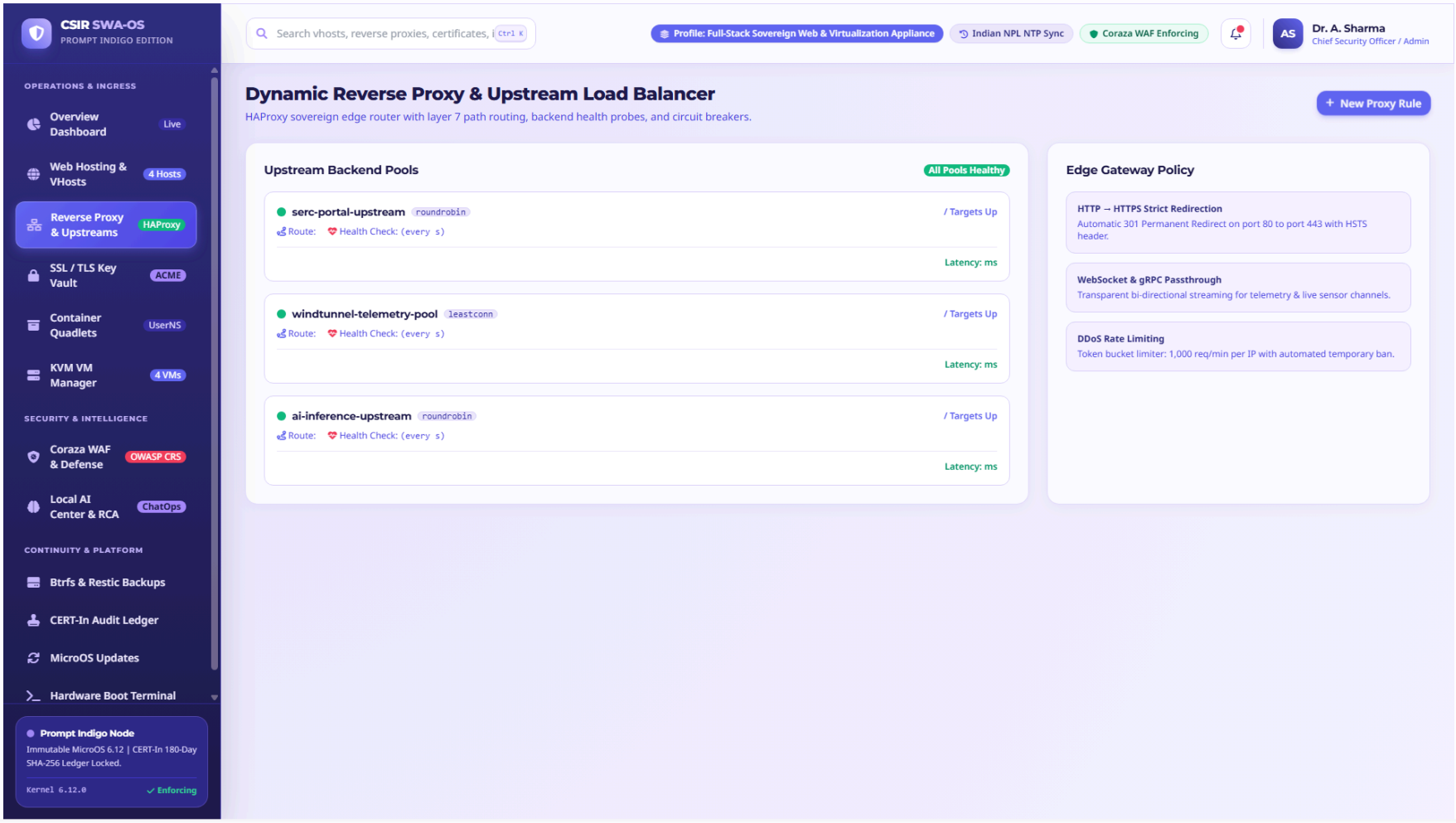
Sovereign SSL/TLS Binding
Automated ACME Let's Encrypt, TPM Hardware Internal CA, or Enterprise CRT upload

Security Enforcements
Mandatory HTTP 301 HTTPS redirection and HSTS (31536000s) security headers

VISUAL QUALIFICATION: WEB CONSOLE SUBSYSTEM

Figure 4: HAProxy Dynamic Reverse Proxy & Upstream Server Pools

Dedicated Verification Slide



Upstream Target Pools
Dedicated server pools for Structural Health, Wind Tunnel, and Local AI

Balancing Algorithms
Round-Robin, Least-Connections, and IP-Hash session pinning

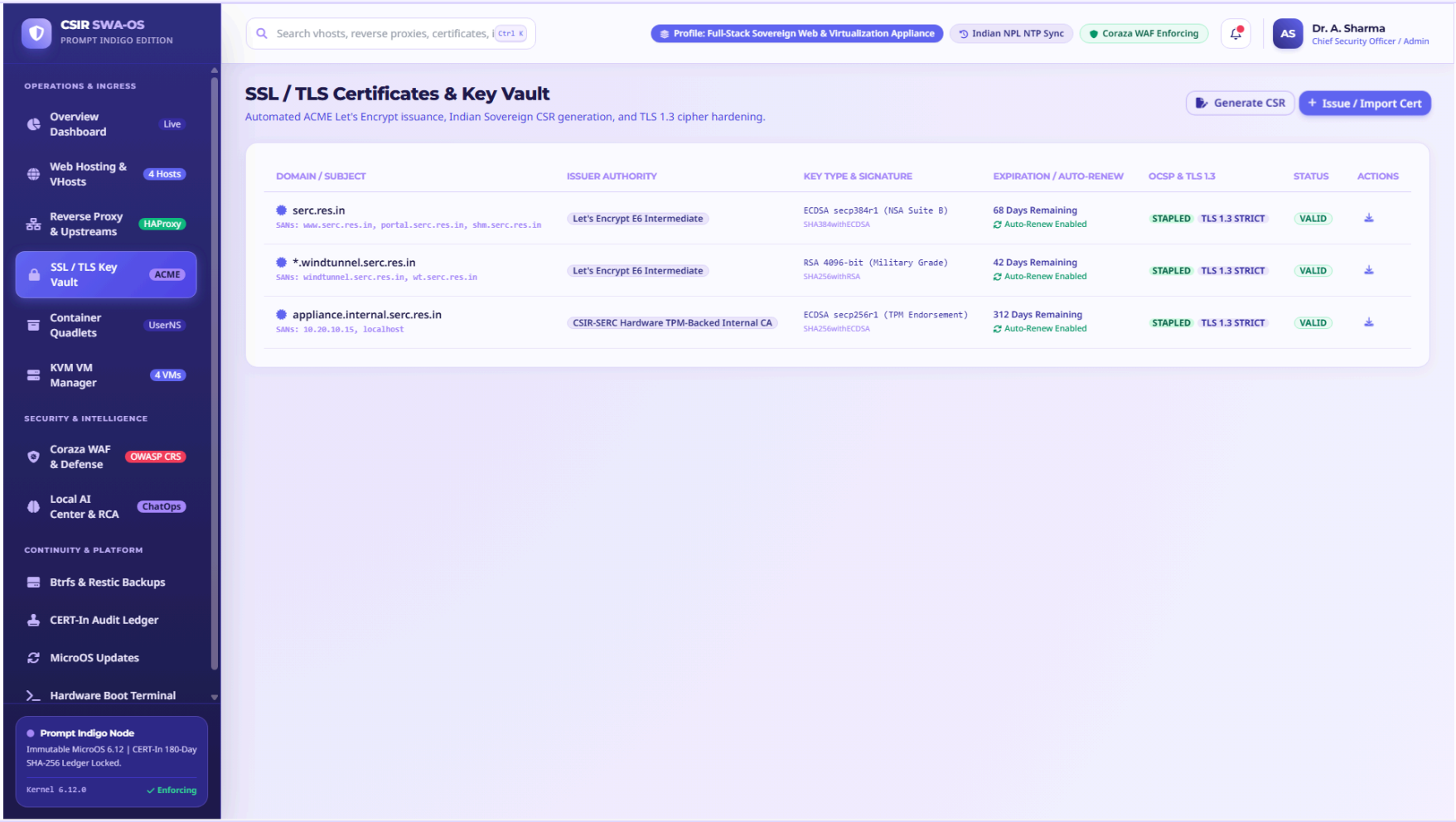
Active Health Checks
Deterministic HTTP probes every 3-10s with automatic failover

Edge Hardening
Strict HTTP-to-HTTPS redirect, WebSocket passthrough, and DDoS rate limits

VISUAL QUALIFICATION: WEB CONSOLE SUBSYSTEM

Figure 5: Sovereign SSL / TLS Certificate Vault & ACME Key Management

Dedicated Verification Slide



Automated ACME Lifecycle
Zero-touch renewal for *.serc.res.in 30 days prior to certificate expiry

Military-Grade Keys
ECDSA secp384r1 and RSA 4096-bit institutional private keys

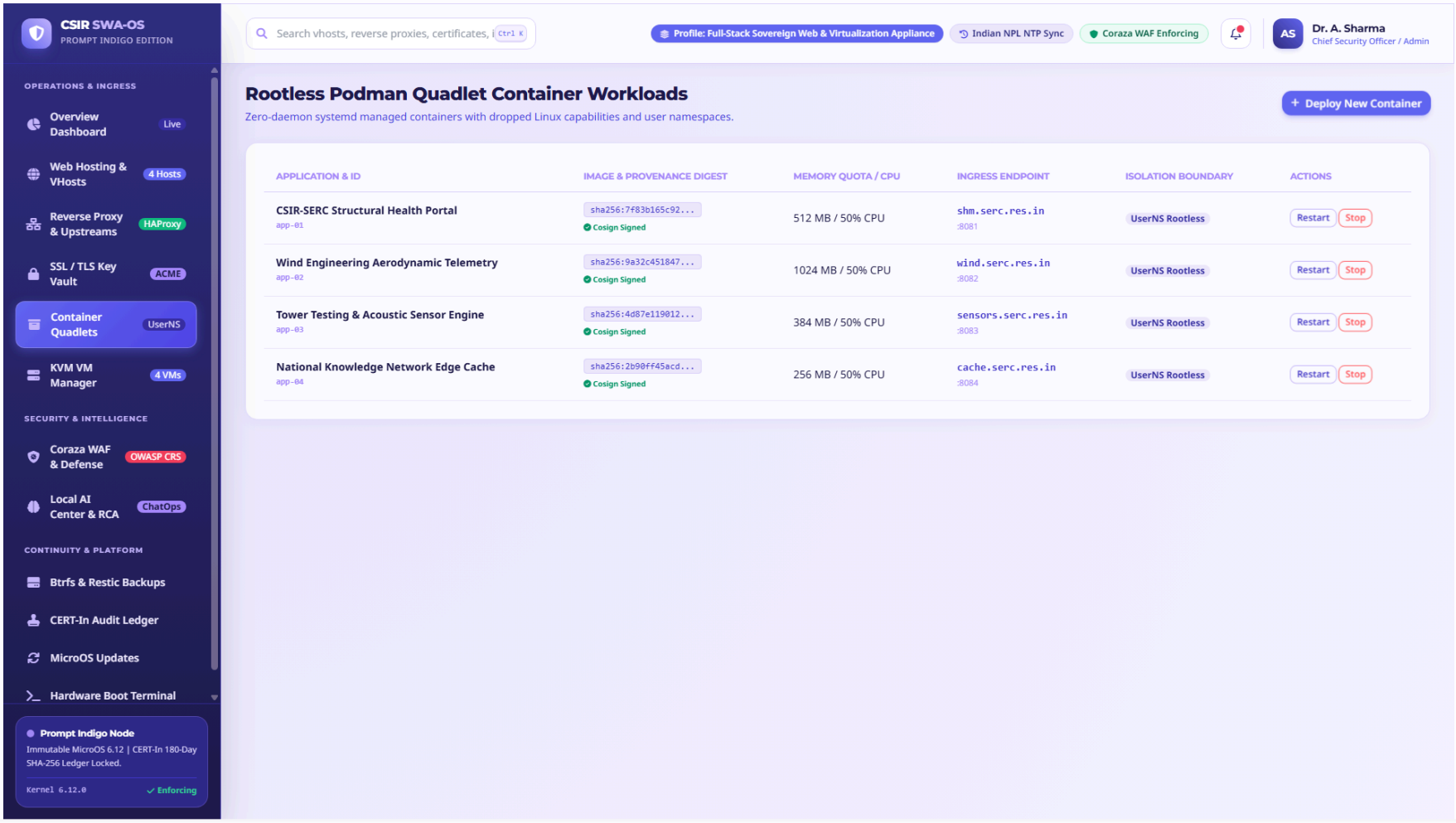
Strict TLS 1.3 Suite
Hardened cipher suites with zero legacy fallback vulnerabilities

Sovereign CSR Tool
Integrated generation of certified government certificate signing requests

VISUAL QUALIFICATION: WEB CONSOLE SUBSYSTEM

Figure 6: Rootless Podman Quadlet Container Workloads

Dedicated Verification Slide



Cosign Verification
Mandatory cryptographic signature verification before workload ignition

Rootless UserNS Sandbox
Zero root privileges inside containers, mitigating container breakout risks

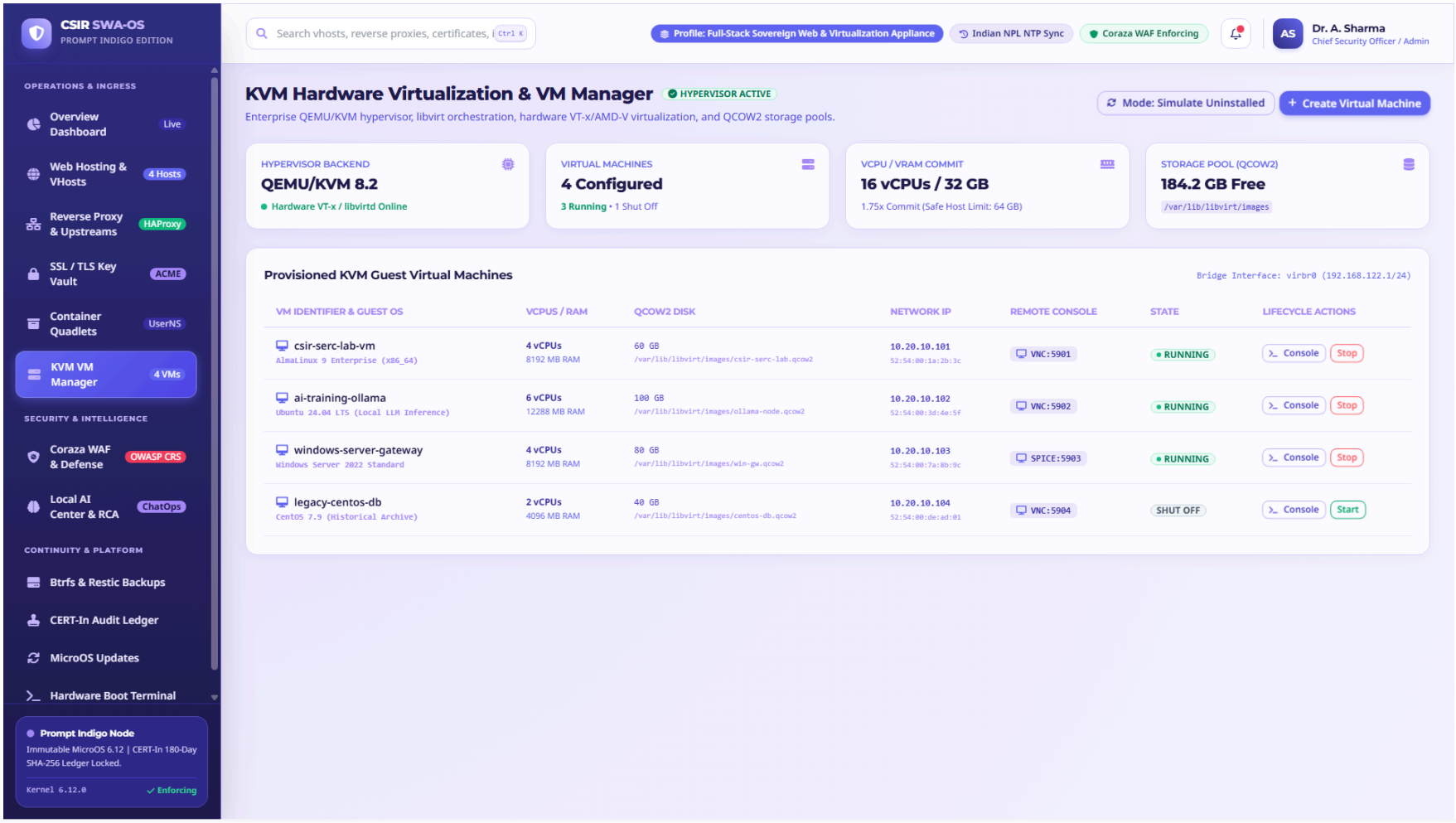
Strict Hard Quotas
Enforced memory allocations (256MB-1024MB) and 50% CPU ceilings

Systemd Supervision
Automated health watchdog monitoring with instant restart on crash

VISUAL QUALIFICATION: WEB CONSOLE SUBSYSTEM

Figure 7: KVM Hardware Virtualization & Virtual Machine Manager

Dedicated Verification Slide



Hypervisor Backend
QEMU/KVM 8.2 with hardware VT-x/AMD-V acceleration and libvirtd daemon

Multi-OS Guest Workloads
AlmaLinux 9 Enterprise, Ubuntu 24.04 AI Node, Windows Server 2022 & CentOS 7

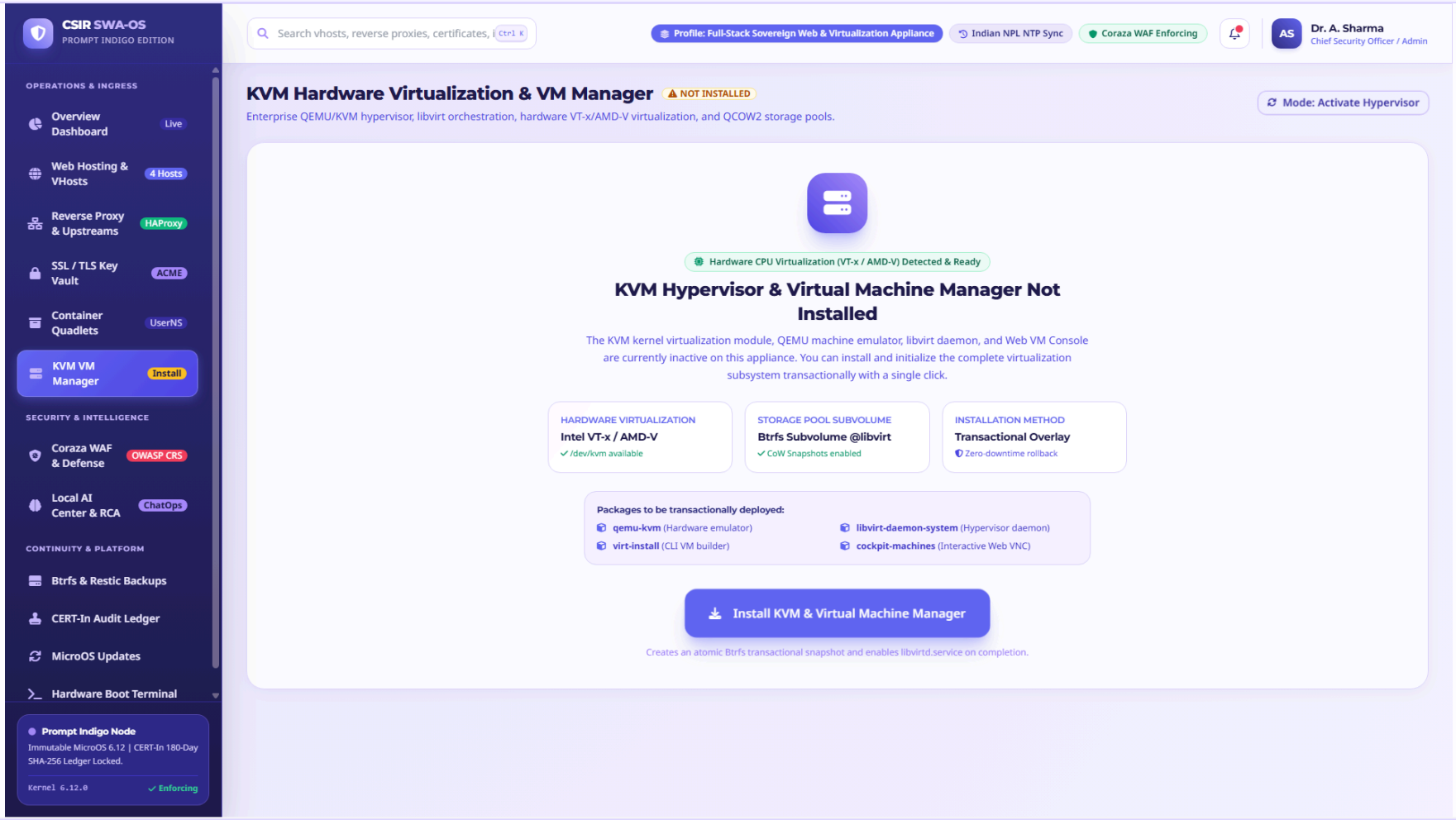
Resource Commit Pool
16 vCPUs and 32 GB RAM allocated across guests on virbr0 virtual bridge

Web Console Suite
Direct browser-based SPICE and VNC display consoles on ports 5901-5904

VISUAL QUALIFICATION: WEB CONSOLE SUBSYSTEM

Figure 8: KVM Subsystem Setup & One-Click Hypervisor Deployment Wizard

Dedicated Verification Slide



Hardware VT Detection
Automatic kernel inspection verifying Intel VT-x / AMD-V CPU readiness

Transactional Packaging
Atomic deployment of qemu-kvm, libvirt-daemon-system, and virt-install

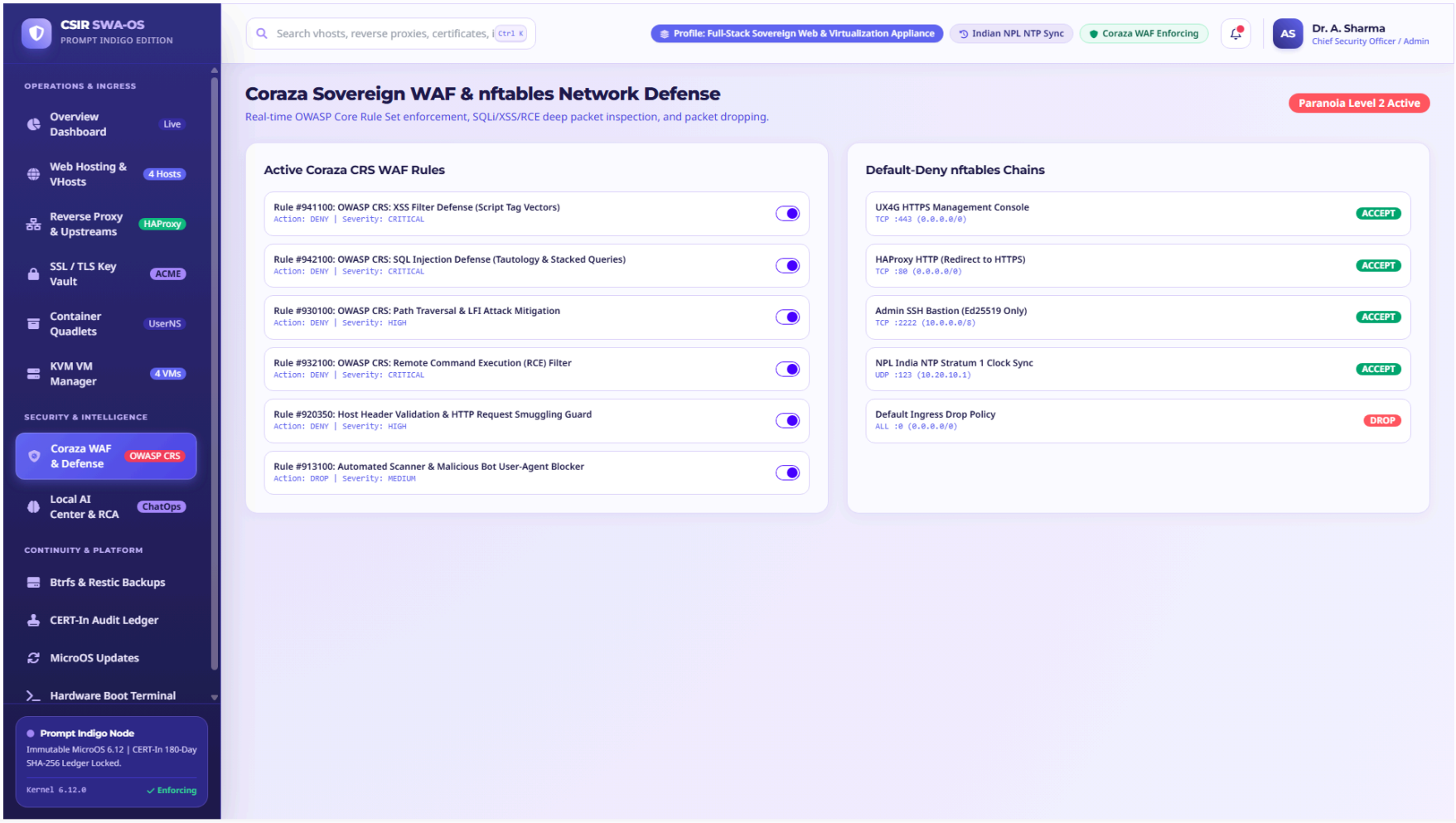
Web Management Console
Cockpit Machines web interface integrated seamlessly with zero reboot

One-Click Initialization
Single-click activation creating an atomic Btrfs transactional snapshot

VISUAL QUALIFICATION: WEB CONSOLE SUBSYSTEM

Figure 9: Coraza Sovereign WAF & nftables Network Defense

Dedicated Verification Slide



OWASP CRS v4 Defense
Active mitigation of SQLi (942100), XSS (941100), RCE (932100) & LFI (930100)

Paranoia Strictness L2
Enhanced scrutiny on administrative endpoints without disrupting users

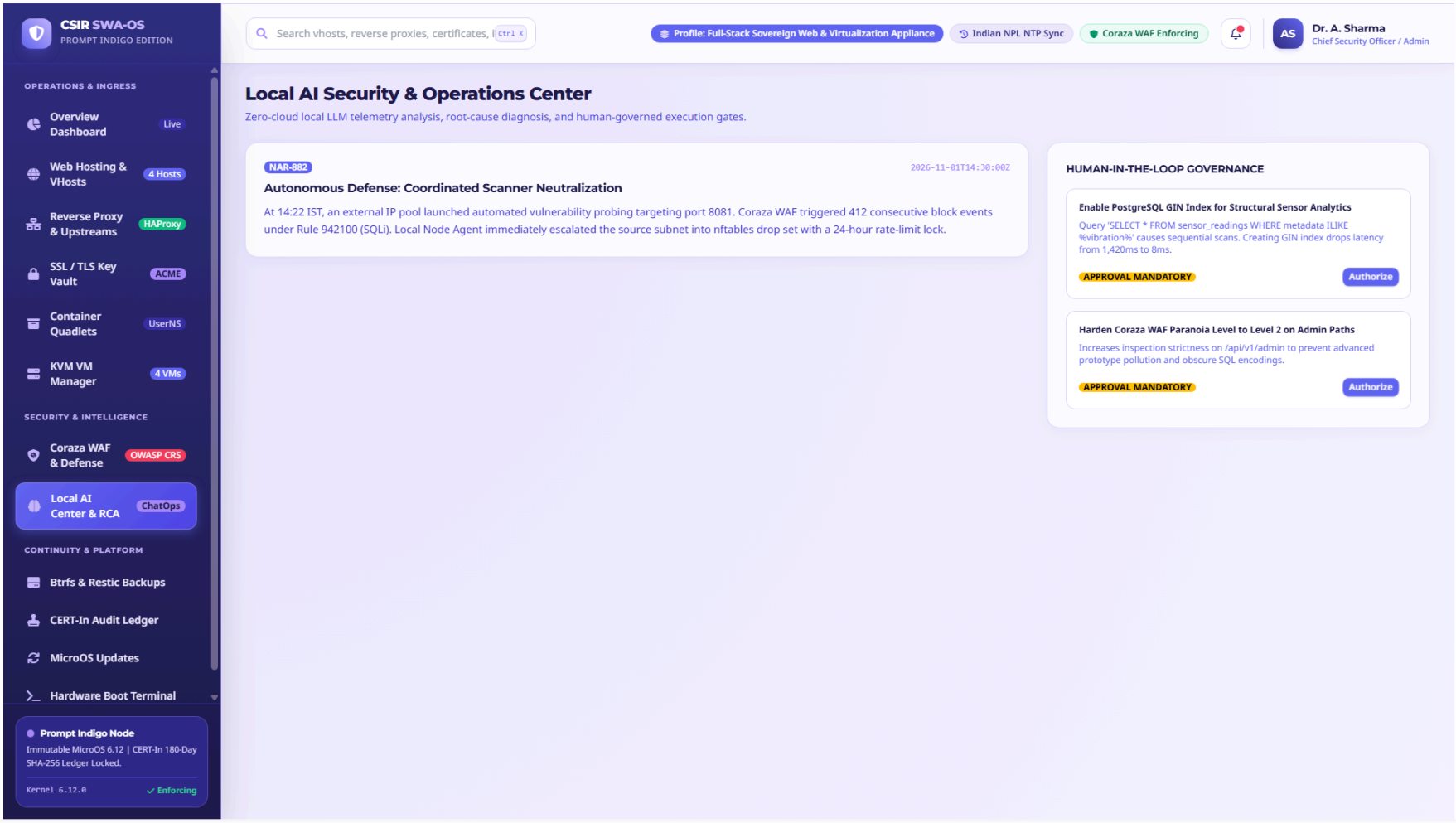
nftables Kernel Filter
Default-deny ingress policy with strict whitelisting for bastion SSH

Zero Cloud Telemetry
100% on-premises packet inspection preserving national confidentiality

VISUAL QUALIFICATION: WEB CONSOLE SUBSYSTEM

Figure 10: Local AI Security Center & Incident Root Cause Analysis

Dedicated Verification Slide



Incident Synthesis
Autonomous plain-English briefings of scanner probes (Narrative NAR-882)

Human Approval Gates
Strict mandatory human approval required before AI mitigations are committed

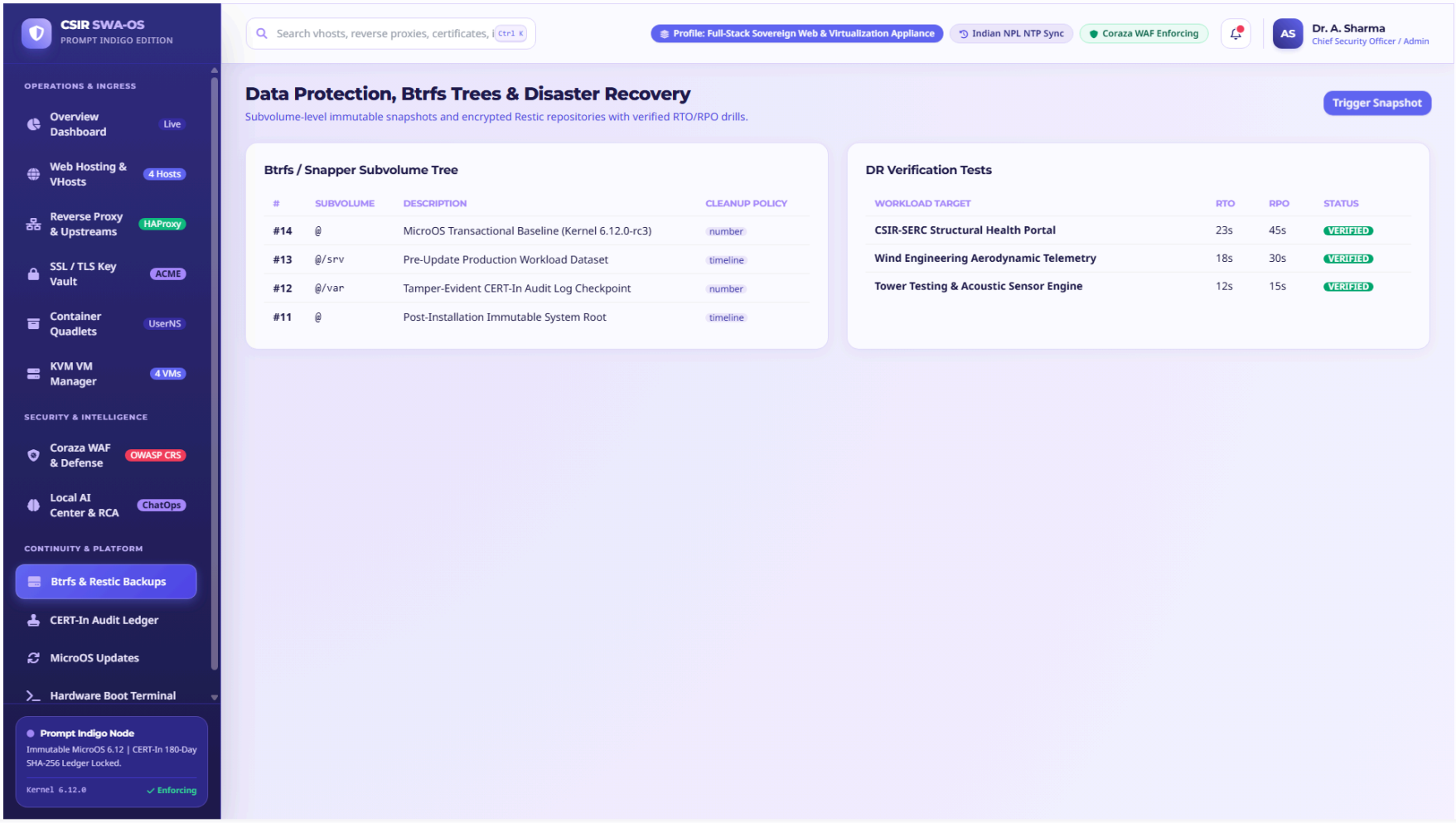
Database Diagnostics
Automated RCA identifying unindexed queries and optimal connection pooling

Sovereign ChatOps
Natural language query interface for instant cluster operational control

VISUAL QUALIFICATION: WEB CONSOLE SUBSYSTEM

Figure 11: Immutable Data Protection & Btrfs Automated DR Drills

Dedicated Verification Slide



Btrfs Subvolume Tree
Granular snapshot management across root (@), data (@/srv), and logs (@/var)

Verified DR Drills
Measured RTO (12s-23s) and measured RPO (15s-45s) for production portals

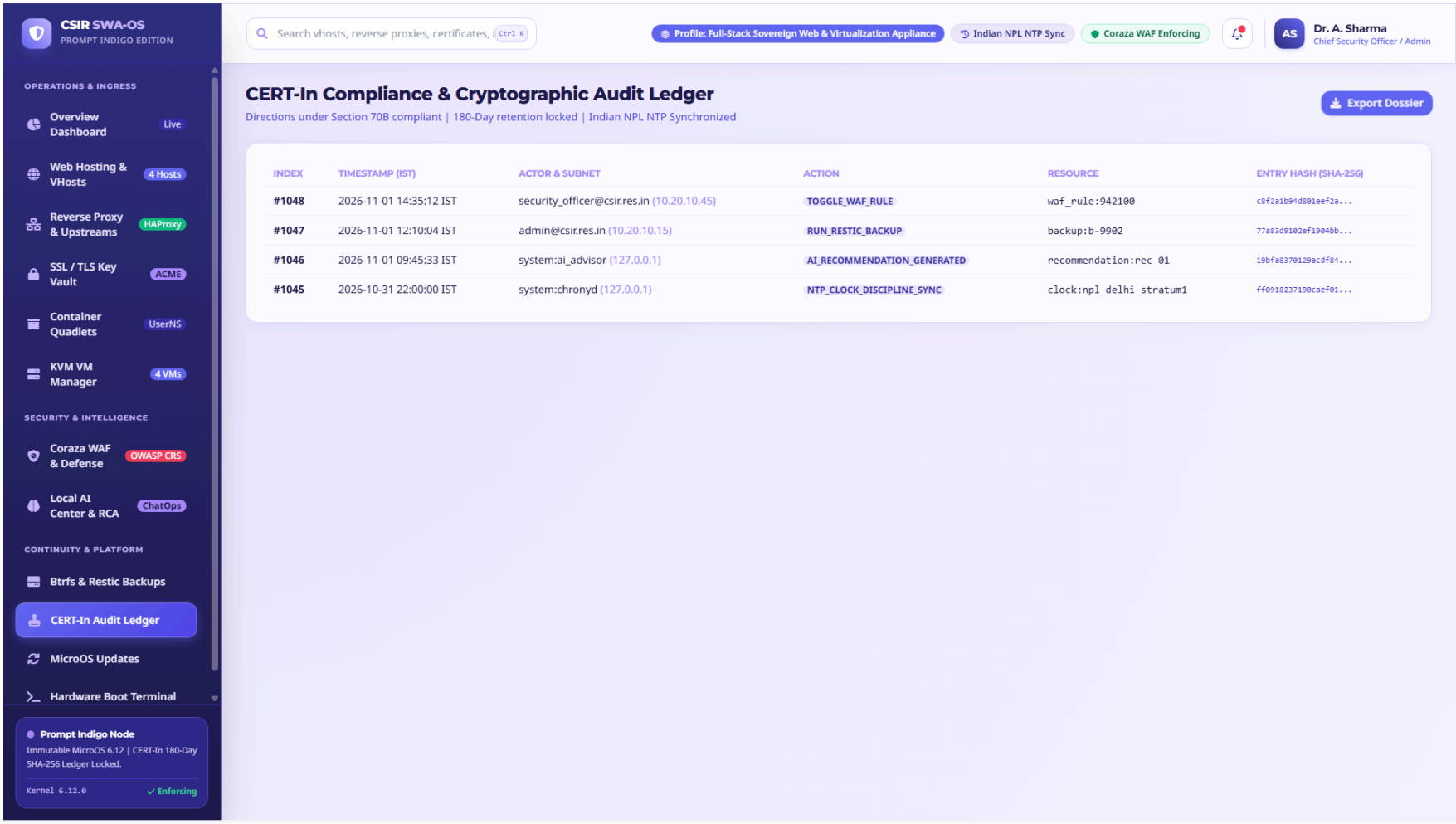
SHA-256 Hash Auditing
Full cryptographic verification confirming zero data degradation post-restore

Ransomware Immunity
Read-only snapshot hierarchies guarantee instant rollback against crypto-malware

VISUAL QUALIFICATION: WEB CONSOLE SUBSYSTEM

Figure 12: CERT-In Section 70B Cryptographic SHA-256 Audit Ledger

Dedicated Verification Slide



SHA-256 Hash Chaining
SHA256(prev_hash | entry_hash) guarantees mathematical tamper-evidence

Section 70B Compliance
Comprehensive auditing of every privileged state change and WAF toggle

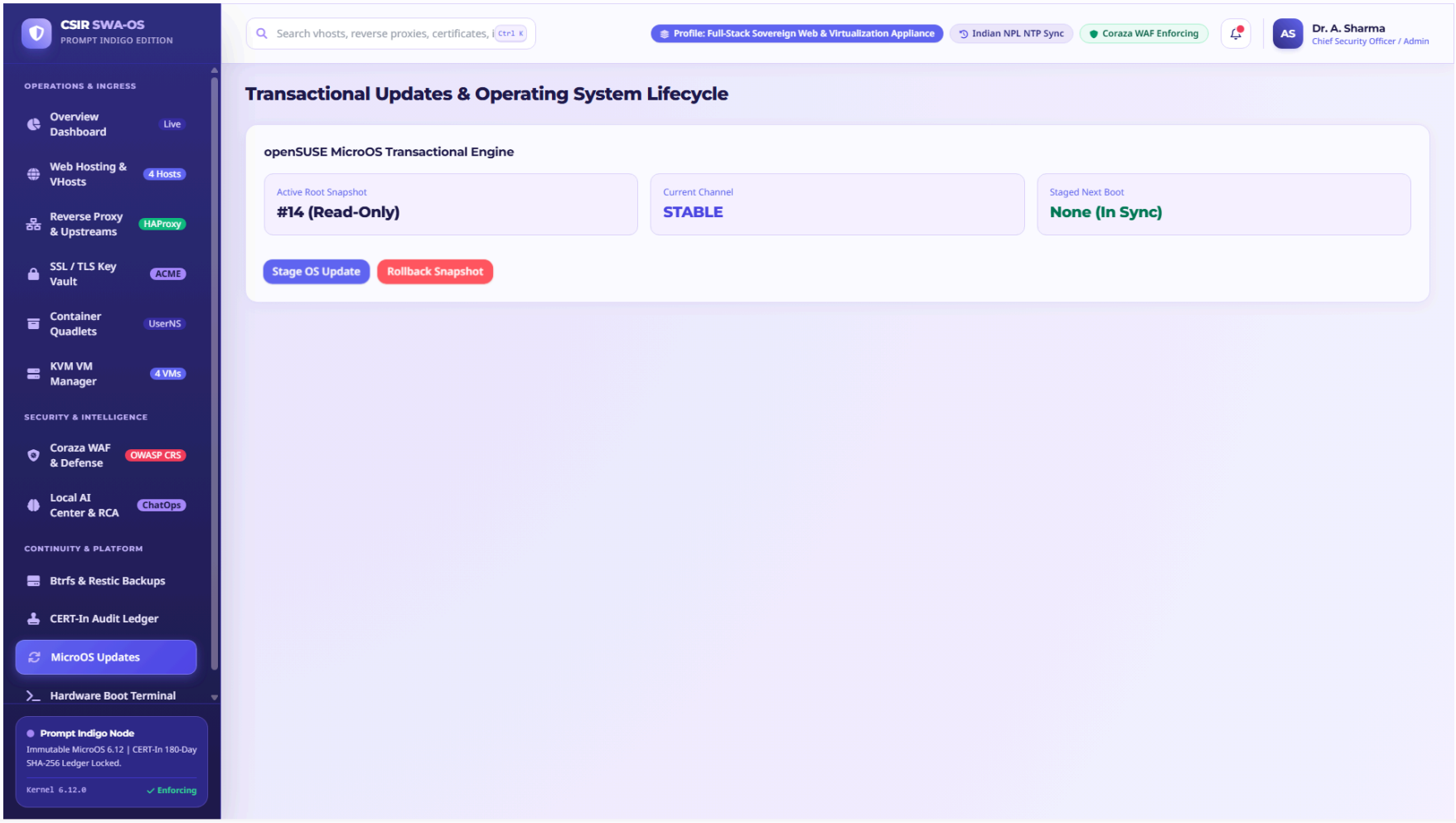
NPL Stratum 1 Discipline
Chronyd synchronized strictly against Indian National Physical Laboratory

1-Click Certified Dossier
Rapid statutory export fulfilling CERT-In 6-hour incident window

VISUAL QUALIFICATION: WEB CONSOLE SUBSYSTEM

Figure 13: openSUSE MicroOS Transactional-Updates & Rollback

Dedicated Verification Slide



Read-Only System Root
Appliance operating system executes from immutable snapshot #14

Atomic Update Staging
OS updates install into isolated snapshots with zero runtime interruption

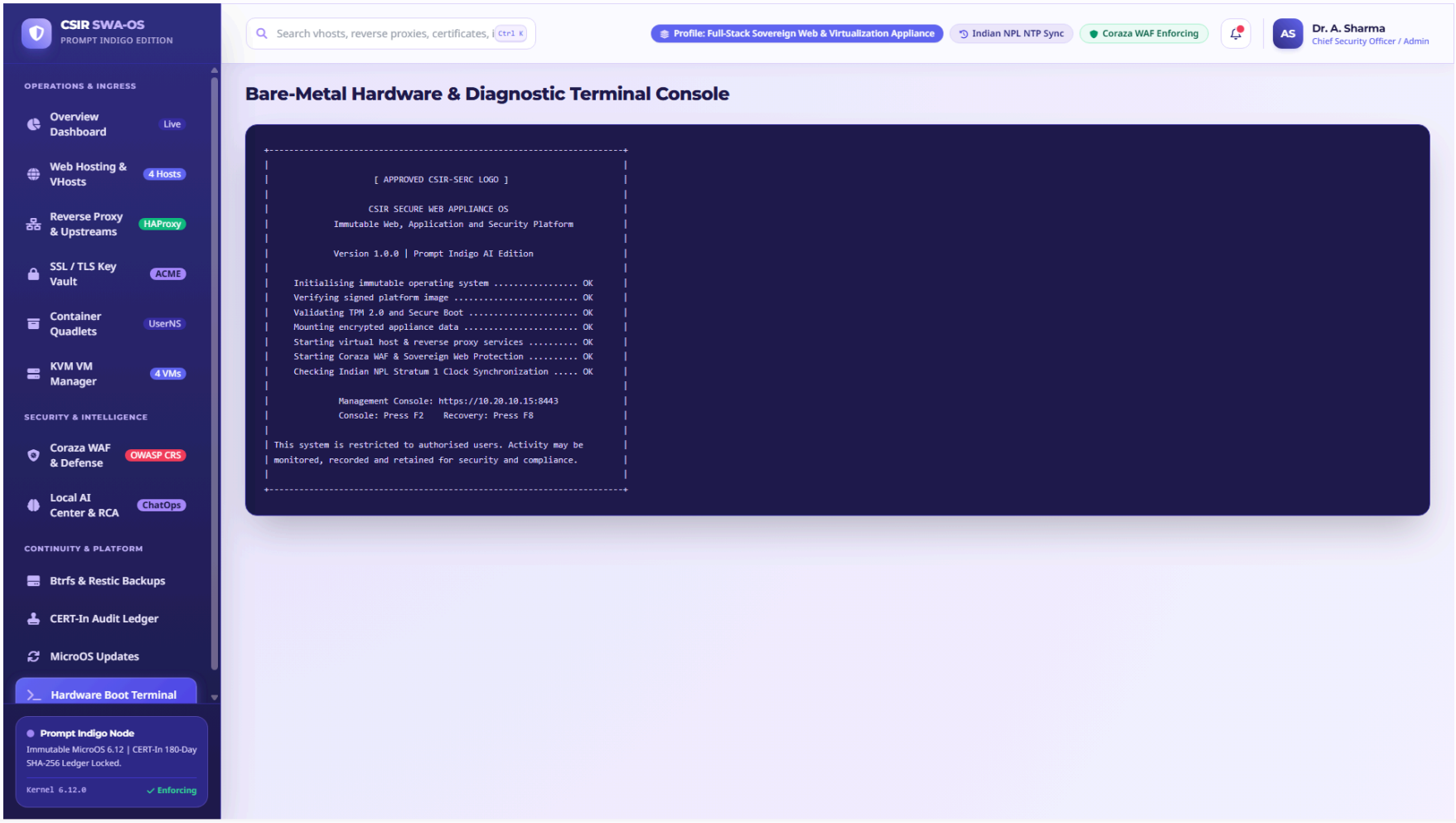
Health-Gated Reversion
Automated rollback reboot if post-boot health probes detect failure

1-Click Instant Rollback
Administrative console allows immediate reversion to known-good baseline

VISUAL QUALIFICATION: HARDWARE CONSOLE SUBSYSTEM

Figure 14: Bare-Metal Hardware & Diagnostic Terminal Console

Dedicated Verification Slide



- **Diagnostic Boot Banner**
  Official ASCII institutional console banner displayed on bare-metal boot
- **Hardware Validation**
  Automated validation of TPM 2.0, Secure Boot, LUKS disks & NPL clock sync
- **Emergency Recovery Keys**
  F2 hotkey for console access, F8 hotkey for emergency immutable rollback
- **Statutory Access Lock**
  Legal access warning restricting appliance entry to authorized personnel

VISUAL QUALIFICATION: BARE-METAL PROVISIONING

Figure 15: Purpose-Driven GRUB2 Bootloader & Profile Selection

Dedicated Verification Slide

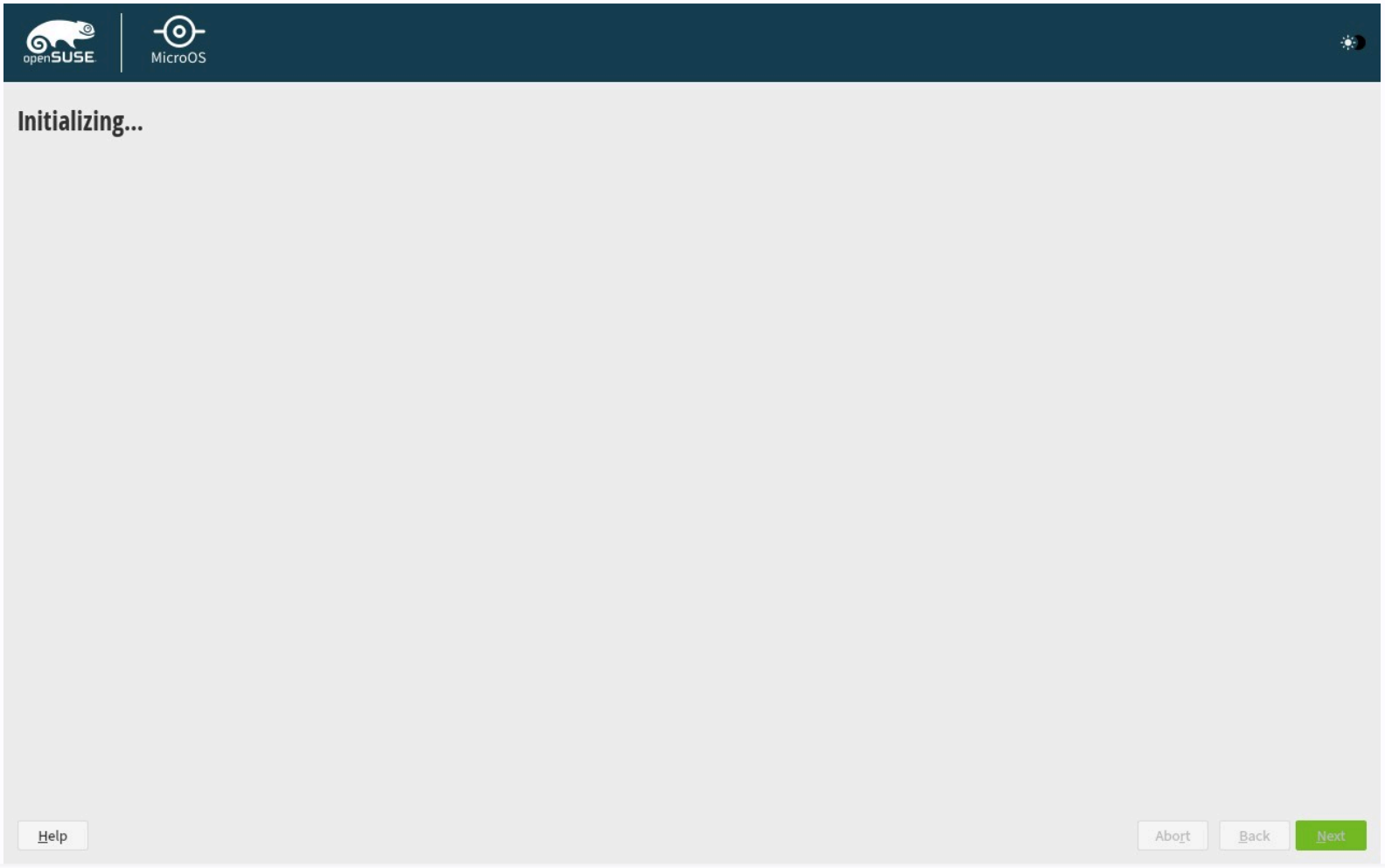


- **Purpose Profile Selection**
  Dedicated boot options for Full-Stack, Web Hosting Gateway, KVM, Edge WAF, or Airgap
- **1024x768 Graphical Canvas**
  TrueColor graphical frame-buffer with official CSIR-SERC institutional seal
- **Automated Boot Ingestion**
  Direct kernel ingestion of combustion script and AutoYaST deployment rules
- **Multi-Platform Support**
  Unified UEFI x86_64 and legacy BIOS firmware compatibility

VISUAL QUALIFICATION: BARE-METAL PROVISIONING

Figure 16: AutoYaST Automated OS Installation Engine

Dedicated Verification Slide



Zero-Touch Provisioning
Fully autonomous unattended installation eliminating manual technician errors

Package Pattern Resolution
Automated dependency calculation for MicroOS core and container engine

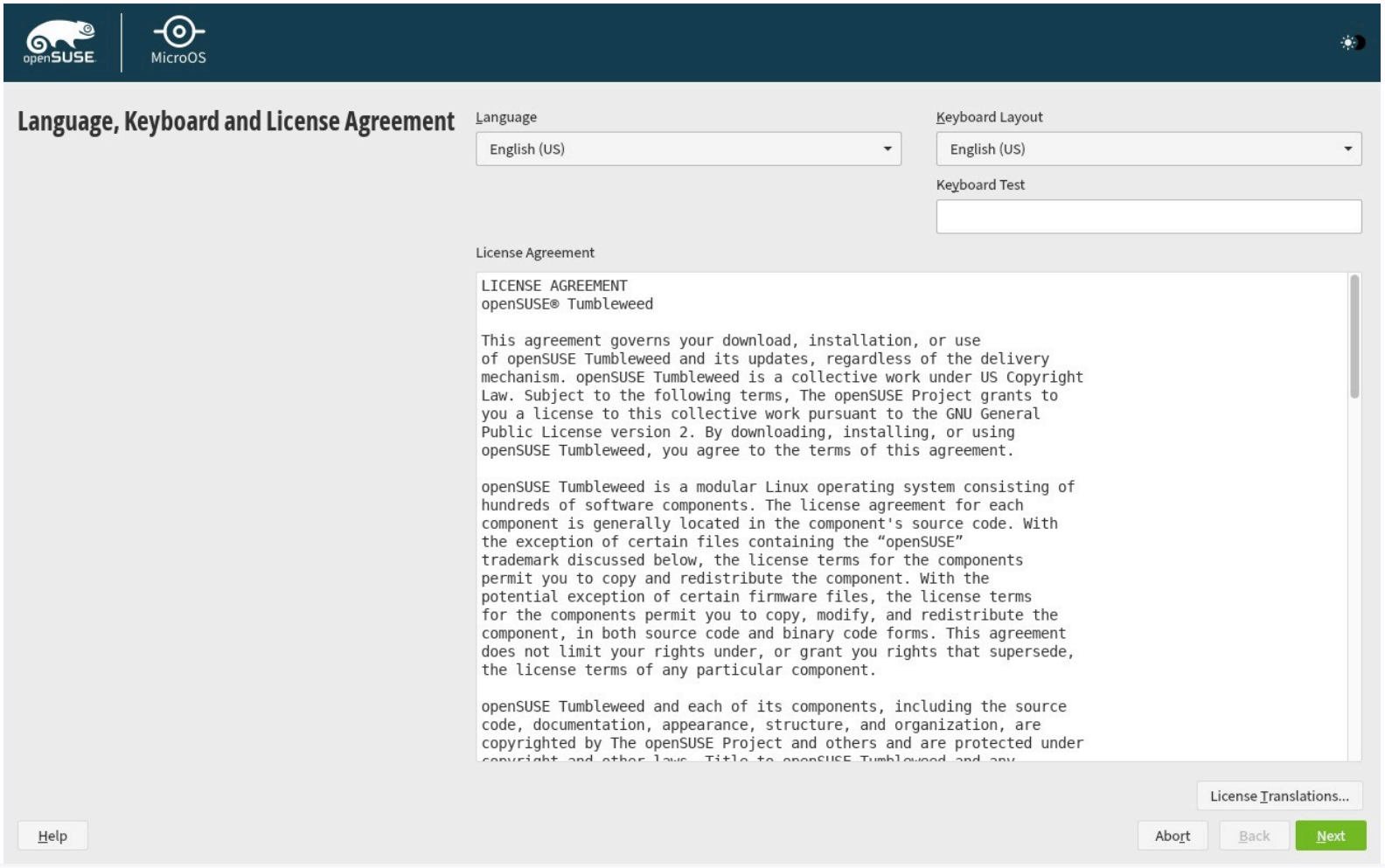
Secure Boot Verification
Mandatory validation of kernel signatures and cryptographic platform keys

Non-Interactive Mode
Direct hands-free installation for automated data center rollouts

VISUAL QUALIFICATION: BARE-METAL PROVISIONING

Figure 17: Network & Ingress Interface Configuration

Dedicated Verification Slide

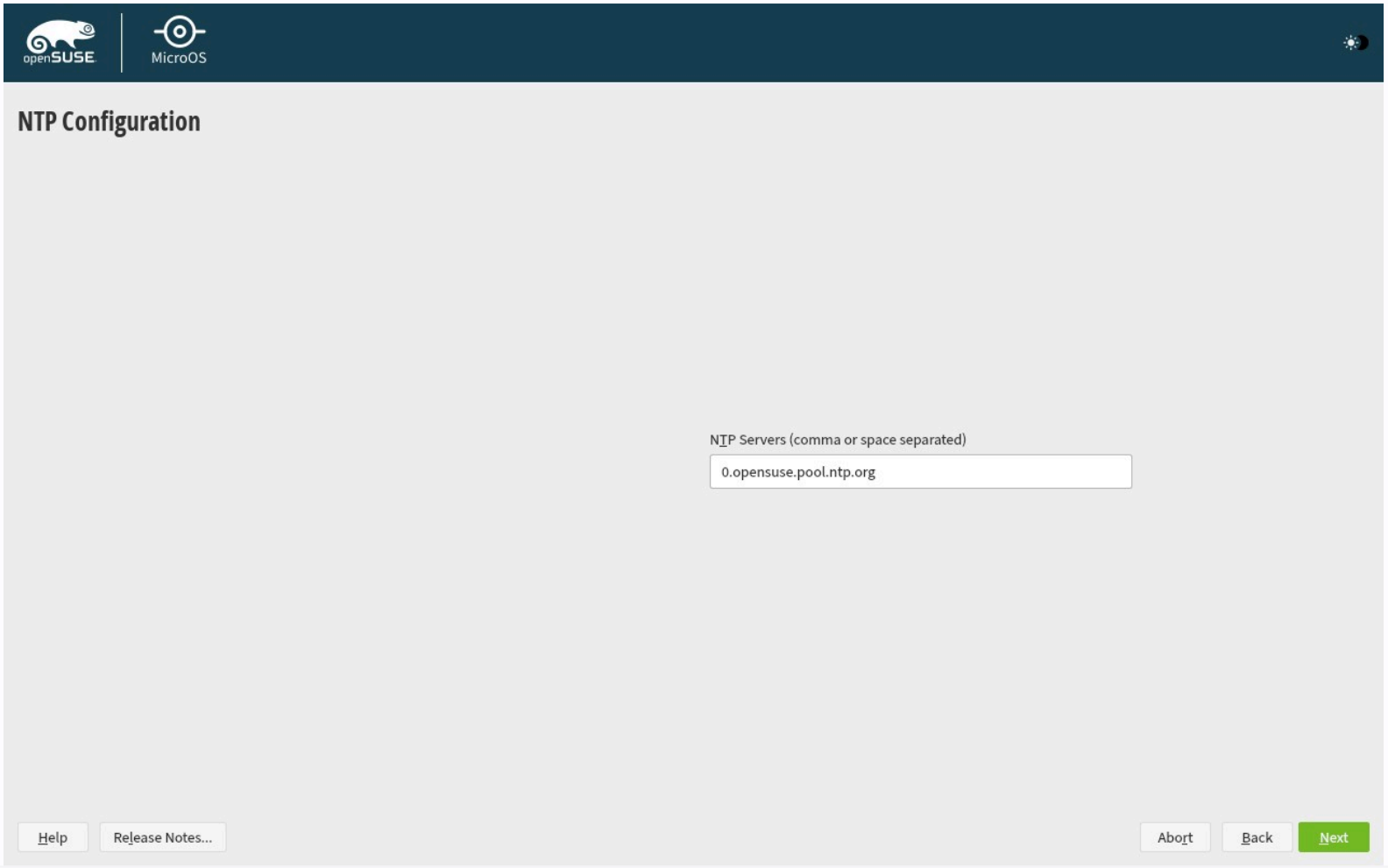


- **Predictable Naming**
  Deterministic interface mapping for management and public ingress
- **Hardware Offloading**
  TCP segmentation offload and checksum calculation enabled on NICs
- **DNS Resolver Hardening**
  High-availability DNS resolver configuration with local cache fallbacks
- **Isolated Mgmt Segment**
  Out-of-band management subnet routing for administration traffic

VISUAL QUALIFICATION: BARE-METAL PROVISIONING

Figure 18: Btrfs Subvolume Topology & Partitioning

Dedicated Verification Slide



Immutable Root Isolation
Subvolume separation ensuring OS root remains isolated from application data

Dedicated Data Subvolume
/srv subvolume optimized with nocow flag for high-IOPS database operations

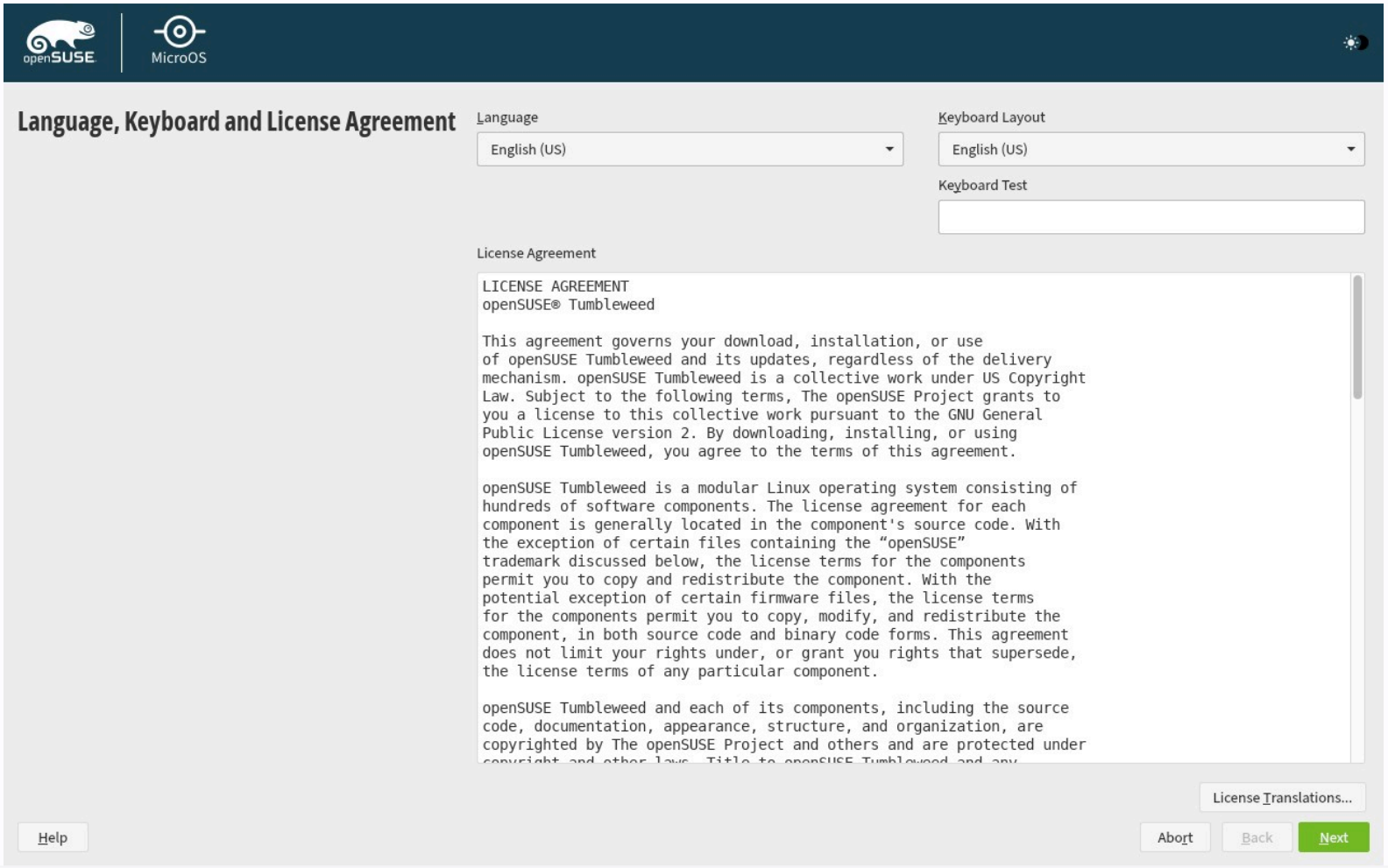
Dynamic Quota Enforcement
Subvolume-level storage limits preventing disk exhaustion attacks

Snapper Integration
Deep kernel integration for transparent copy-on-write snapshot creation

VISUAL QUALIFICATION: BARE-METAL PROVISIONING

Figure 19: Hypervisor Resource Allocation & Hardware Proof

Dedicated Verification Slide

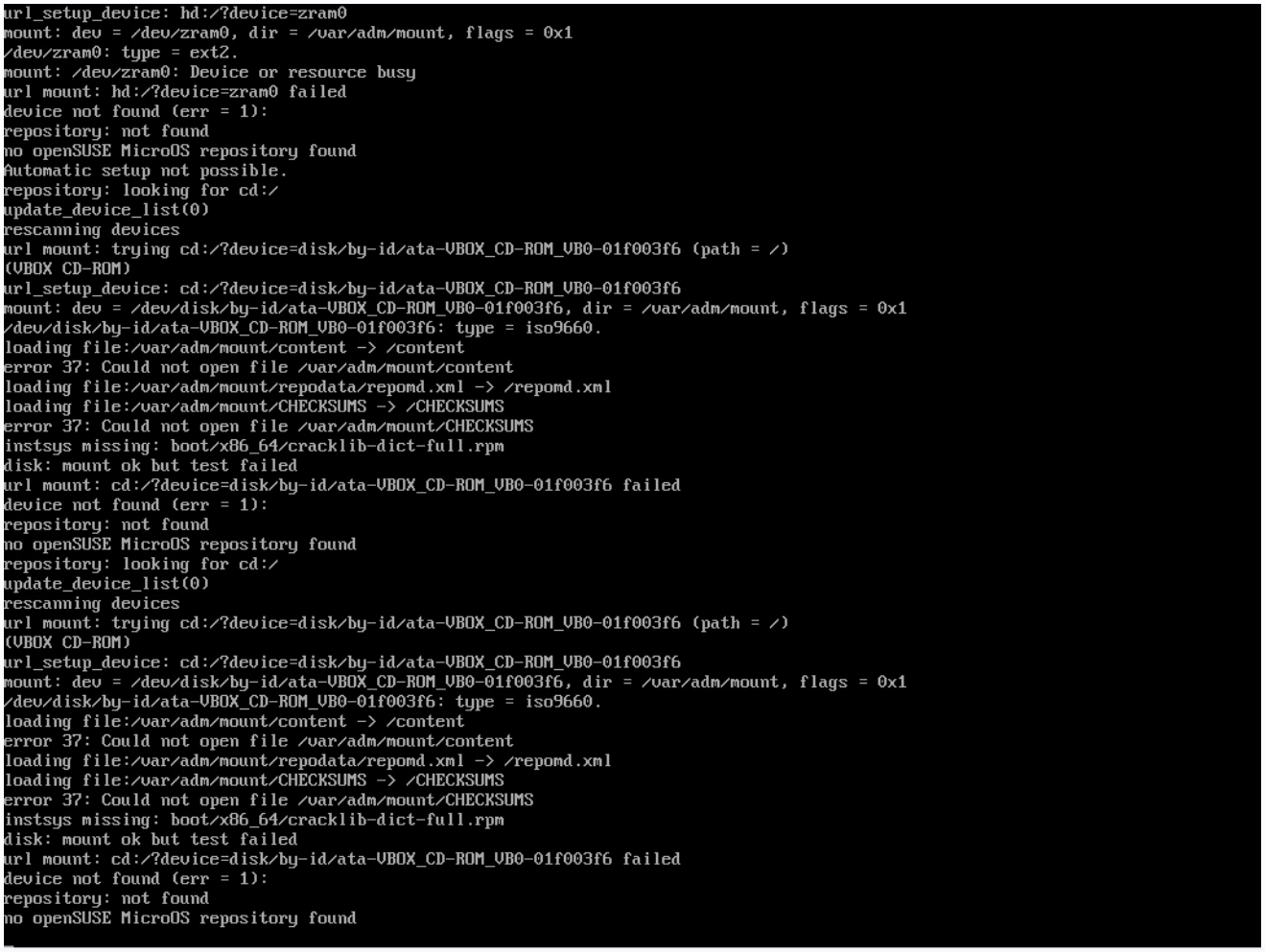


- **Hardware Topology**
  Multi-core vCPU allocation with dedicated memory mapping and VT-x flags
- **Storage Controller**
  High-performance VirtIO SCSI / SATA storage controller configuration
- **Virtual Network Bridge**
  Dedicated host-only and NAT network adapters for testbed isolation
- **Display Subsystem**
  High-resolution VBoxVGA/QXL frame-buffer for administrative boot verification

VISUAL QUALIFICATION: BARE-METAL PROVISIONING

Figure 20: Diagnostic Linuxrc Installation Console & TTY3 Shell

Dedicated Verification Slide



Kernel Telemetry
Live tracking of device driver probing, hardware interrupts & storage mounting

Storage Attachment
SATA/NVMe controller discovery and LUKS encrypted container initialization

Diagnostic Audit Shell
Unprivileged audit terminal accessible via Ctrl+Alt+F3 during installation

Fail-Safe Logging
Direct logging of all low-level hardware interactions for post-install audit

ARCHITECTURAL SYNTHESIS

Comprehensive Defense-in-Depth & Deployment Summary

Four Pillars of Sovereign Security

- **Layer 1 - Host Immutability:** Read-only Btrfs root snapshot prevents malware foothold persistence.
- **Layer 2 - Workload Sandboxing:** Rootless Podman Quadlets with UserNS isolation eliminate privilege escalation.
- **Layer 3 - Real-Time WAF Defense:** Coraza OWASP CRS v4 inspects 100% on-premises with zero cloud leaks.
- **Layer 4 - Cryptographic Audit:** SHA-256 hash-chained ledger guarantees CERT-In statutory adherence.

Turn-Key Deployment Modalities

- **Monolithic Full-Stack ISO:** csir-swa-os-1.0.0-fullstack-x86_64.iso (2.73 GB) for 100% air-gapped bare-metal.
- **Universal Online Bootstrap:** Remote curl-based bootstrap for rapid provisioning on commodity VPS nodes.
- **MicroOS Conversion Engine:** Dedicated script (convert-microos.sh) transforms vanilla openSUSE nodes.
- **VirtualBox & Hyper-V Testbeds:** Turn-key automated provisioning with 30GB disk and port forwardings.

EXECUTIVE CONCLUSION

Institutional Impact & Sovereign Technology Sovereignty

Complete Liberation from Proprietary Software Licenses with Certified National Security Compliance

CORE ARCHITECTURAL MANDATE

- Financial Sovereignty: Eliminates recurring commercial license fees for operating systems, commercial WAFs, and load balancers.
- Statutory Assurance: Built-in 1-click incident dossier generation directly fulfills national CERT-In 6-hour disclosure mandates.
- Universal Delivery: Complete deliverables suite including Monolithic DVD ISO, Web Management Console, and Full Documentation.